Population Studies
A Journal of Demography

ISSN: 0032-4728 (Print) 1477-4747 (Online) Journal homepage: www.tandfonline.com/journals/rpst20

# Measuring age differences among different-sex couples: Across religions and 130 countries, men are older than their female partners

Jacob Ausubel, Stephanie Kramer, Anne Fengyan Shi & Conrad Hackett

**To cite this article:** Jacob Ausubel, Stephanie Kramer, Anne Fengyan Shi & Conrad Hackett (2022) Measuring age differences among different-sex couples: Across religions and 130 countries, men are older than their female partners, Population Studies, 76:3, 465-476, DOI: 10.1080/00324728.2022.2094452

**To link to this article:** https://doi.org/10.1080/00324728.2022.2094452

View supplementary material

Published online: 27 Sep 2022.

Submit your article to this journal

Article views: 855

View related articles

View Crossmark data

Citing articles: 11 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=rpst20

*Population Studies*, 2022
Vol. 76, No. 3, 465–476, https://doi.org/10.1080/00324728.2022.2094452

# Measuring age differences among different-sex couples: Across religions and 130 countries, men are older than their female partners

**Jacob Ausubel [1], Stephanie Kramer [1], Anne Fengyan Shi [1] and Conrad Hackett [1,2]**
[1]Pew Research Center, [2]University of Maryland

*Cross-national studies of age patterns among couples have tended to compare the ages at which men and women first marry, but few have analysed age differences between current spouses or cohabiting partners (i.e. partner age gaps). We address this gap in the literature by using recent census and survey data to analyse age differences between current partners in 130 countries. Worldwide, men are an average of 4.2 years older than their wives or cohabiting partners. However, age gaps vary by region, ranging from 8.6 years in sub-Saharan Africa to 2.2 years in North America. The religious group with the largest age gap is Muslims (6.6 years), while the smallest age gap is seen among Jews (2.2 years). Differences between Muslims and non-Muslims remain even after controlling for country-level gender inequality and per-capita GDP.*

Supplementary material is available for this article at: https://doi.org/10.1080/00324728.2022.2094452

**Keywords**: age differences; partners; religion; singulate mean age at marriage; age-heterogamous marriages

[Submitted November 2021; Final version accepted May 2022]

## Introduction

Cross-national studies of age patterns among couples have tended to compare the ages at which men and women first marry, but few have analysed age differences between current spouses or cohabiting partners (i.e. partner age gaps). We address this gap in the literature by using recent census and survey data to analyse age differences between current partners in 130 countries, which cover 91 per cent of the global population. The data sets we analyse comprise data from 82.6 million individuals in 20.8 million households.

Our rationale for analysing unmarried partners alongside formally married spouses is that cohabitation is an increasingly common living arrangement in Europe, Latin America and the Caribbean, and North America. This is especially true among young adults, who typically cohabit as a precursor or alternative to marriage (OECD 2011; United Nations 2019). Also, while some data sources differentiate a married spouse from a cohabiter, many do not. Looking at ages of current spouses or partners reveals slightly different patterns from considering age at first marriage. A large age differences between partners is more common among women and men who remarry than among first-time brides and grooms (Shehan et al. 1991; Berardo et al. 1993; Wu et al. 2000; Conaway 2002; Livingston 2014). Our study focuses on different-sex couples, as data on same-sex couples are frequently unavailable, particularly in countries where same-sex marriage remains illegal (BBC 2021).

We find that worldwide, men are an average of 4.2 years older than their wives or female cohabiting partners. However, mean age gaps vary by region, ranging from 8.6 years in sub-Saharan Africa to 2.2 years in North America. The religious group with the largest age gap is Muslims (6.6 years), while the smallest age gap is seen among Jews (2.2 years). Global differences between Muslims and non-

© Pew Research Center 2022. All Rights Reserved.

Muslims remain even after controlling for country-level gender inequality and per-capita GDP.

We also consider differences in the singulate mean age at marriage (SMAM) by sex. Unlike the partner age gap, which is the age difference between current partners, the SMAM difference measures the distance between the ages at which men and women first marry. In an analysis of 118 countries where recent data are available for both measures, we find that on average men are 3.3 years older than women at the time of first marriage (the SMAM difference), whereas in terms of age differences of current partners, men are 4.2 years older on average than their female spouses or cohabiting partners. The two measures are furthest apart in sub-Saharan Africa and in the Middle East and North Africa (MENA).

## Background

### *Definitions*

For our purposes, the word 'partner' refers both to individuals who are married and those who are cohabiting with a partner. The 'partner age gap' refers to the difference in age between the male and female partners in a marital or cohabiting union.

The SMAM is the 'average length of single life expressed in years among those who marry before age 50' (United Nations 2013, p. 1). The SMAM is computed by using the proportions of people who are single in five-year birth cohorts; it can be calculated for men and women separately. Only men and women aged 15–54 are included in the calculations (Hajnal 1953; United Nations 1983, 2013). The SMAM difference measures the distance between the ages at which men and women first marry.

Other age measures referenced in the paper are the mean age at first marriage, referring to the average age of a person when they first get married (United Nations 2013), and median age at first marriage, defined as the age at which half of all men or women have entered their first marriage (Singh and Samara 1996).

Age-heterogamous marriages—sometimes also referred to as 'May–December' marriages or age-dissimilar marriages—are marriages in which there is a large age difference between spouses. Men sometimes marry older women: a famous example is French President Emmanuel Macron, whose wife, Brigitte Macron, is nearly 25 years his senior (Caplan 2017). However, it is much more common for men to marry younger women (Vera et al. 1985; Berardo et al. 1993), as in the case of former United States (US) President Donald Trump and his third wife, Melania Trump, who is about 24 years his junior. The age gaps in these presidential unions are much larger than the average partner age gaps in their respective countries (2.3 years in France, 2.2 years in the US; both in the direction of men being older, on average, than their female partners). We find no country in which, on average, women are older than their male spouses or partners.

### *Underlying factors explaining age patterns*

Over the past half century, sociologists have conducted many studies analysing age differences between spouses and the ages at which men and women marry. Although some studies focused on a single country and others were cross-national, a few common patterns have emerged.

First, education is tied to when men and women marry. In countries with higher levels of education, women tend to marry at older ages and there are smaller gaps between the ages at which men and women first marry (Garenne 2004; Saardchom and Lemaire 2005; Carmichael 2011). These patterns seem to square with Gary S. Becker's economic theory of marriage, in which men and women marry young and universally if household roles are rigidly determined by sex, such that women work in the household while men generate the family income (Becker 1974, 1981; Carmichael 2011; Allendorf et al. 2017). However, the more highly women are educated, the more likely they are to marry later, if at all, as they can rely on their increased earning power (Carmichael 2011).

A second factor linked with age at marriage is income. In OECD countries, such as the US and Sweden, age-heterogamous marriages are more prevalent among people with lower incomes (Atkinson and Glass 1985; Vera et al. 1985; Gustafson and Fransson 2015). The same is true for countries in sub-Saharan Africa (Garenne 2004). Higher income levels are associated with more egalitarian attitudes towards gender roles, and smaller age discrepancies between spouses are usually seen in societies with greater gender equality(Atkinson and Glass 1985).

Factors associated with larger age gaps include remarriage and younger brides. Age differences tend to be larger among women and men who remarry compared with first-time brides and grooms (Shehan et al. 1991; Berardo et al. 1993; Wu et al. 2000; Conaway 2002; Livingston 2014).

The younger the age of first-time brides, the larger the age gap with their husbands tends to be (MacQuarrie 2016; United Nations 2016). This is especially true of those who marry as children. Child marriage is most common in West and Central Africa, in low-income countries with high levels of poverty, and among Muslims (Wodon 2015; Kohno et al. 2020; UNPF 2020).

Partner age gaps tend to be larger in more patriarchal societies, especially patrilocal societies in which a married couple settles with or near the husband's family (Casterline et al. 1986). Wives who are younger than their husbands may be perceived as more 'easily molded into the roles desired by the husband's family' (Casterline et al. 1986, p. 369).

Religion is associated with differences in the age at which people marry. In countries with higher shares of Muslims, first-time brides tend to marry at younger ages (Garenne 2004). Such countries also tend to report larger gaps, on average, between the ages at which men and women first marry (Carmichael 2011). Of course, these patterns may reflect historical, political, or economic circumstances—including relatively low levels of education and income in some countries with large Muslim populations—rather than specifically *religious* customs, traditions, or teachings.

### *Importance of age differences between partners*

The pattern of women being younger than their male partners shapes living arrangements across the world. Since women tend to be younger than their male partners and also to live longer, older women are disproportionately likely to live alone. Globally, one in five women aged 60 and older lives in a solo household, compared with only 11 per cent of men in the same age range. By contrast, older men are more likely than older women (36 vs 27 per cent) to live with a spouse or cohabiting partner (and no other household members) (Kramer et al. 2019).

The age gap also has implications for women's well-being. Women who are married before age 18, and whose age gaps with their male partners tend to be large, are especially likely to become pregnant before reaching sexual maturity (UNPF 2020). Child brides are also vulnerable to sexually transmitted infections, enduring poverty, and abusive relationships in which they lack the ability to advocate for themselves (UNPF 2020). In addition, some studies have suggested that couples with large age differences are less likely to use contraception (Barbieri et al. 2005).

## Data and methods

### *Age differences between current partners*

We analyse partner age gaps in 130 countries and territories. Twenty-six of the countries are in the Asia-Pacific region, 35 in Europe, 19 in Latin America and the Caribbean, eight in the MENA region, two in North America, and 40 in sub-Saharan Africa. Collectively, these countries are home to 91 per cent of the global population. The data sets we analyse for the 130 countries comprise information on 82.6 million individuals in 20.8 million households. Where possible, country-level data come from IPUMS census samples. Other data sources include the Demographic and Health Survey (DHS), Multiple Indicator Cluster Survey (MICS), and European Social Survey (ESS). Table A1 in the supplementary material lists the source used for each country, as well as sample sizes.

This study disaggregates partner age gaps by religious group. Six groups are included in this analysis: in decreasing order of estimated population size, globally, these are Christians, Muslims, the religiously unaffiliated, Hindus, Buddhists, and Jews. Generally speaking, sample sizes are sufficient for reliably analysing data on the largest religious groups in each country. However, the global size and geographic distribution of religious groups varies greatly (Hackett et al. 2012). Partner age gap data are available for Christians in 113 countries, Muslims in 74 countries, the unaffiliated in 84 countries, Hindus in 12 countries, Buddhists in 18 countries, and Jews in six countries (see Table 1). Individuals affiliated with other religious groups are included in the overall results but are not reported on separately. Properly classifying survey information about religion can be complex (Hackett 2014). This project benefits from the research team's extensive experience in categorizing census and survey data on religion from around the

**Table 1** Number of countries in which partner age gap data are available by religious group

| Religious group | Number of countries |
|---|---|
| Christians | 113 |
| Muslims | 74 |
| Unaffiliated | 84 |
| Hindus | 12 |
| Buddhists | 18 |
| Jews | 6 |

*Source*: 2010–18 census and survey data; see Table A1, supplementary material for details.

world from previous demographic studies (Hackett et al. 2015, 2016, 2017, 2018, 2019, Kramer et al. 2019).

Every source includes a household roster in which each household member's relationship with the household head or survey respondent is recorded. Couples are identified as respondents plus household members who are identified as their spouse or cohabiting partner. Response options describing relationships within households vary across surveys. For example, the DHS typically lists 'wife or husband' as a response option, whereas the ESS offers a broader 'husband/wife/partner' option. Question wording and response options for each survey are available in Table B1 (supplementary material).

Occasionally, we identify subfamily couples, that is, married couples residing in the same household as the primary couple (note that the primary couple consists of the household head and their partner). Subfamily couples can include a household head's parents, children and their spouses, other relatives, or non-relatives. Such couples are excluded from the analysis because they cannot be matched with certainty. For further details, see Table C1 (supplementary material).

Couples' religion is based on the religious affiliation of the household head or respondent to surveys. Globally, interfaith unions appear to be rare, and some data sets do not reveal the religious affiliations of all household members. For countries in which at least 95 per cent of the population is estimated to belong to the same religious group (based on Pew Research Center's unpublished 2015 religious composition estimates), the predominant religious affiliation is applied to all couples.

The age gap for each couple is calculated by subtracting the female partner's age in years from the male partner's age. Mean age gaps are then calculated across countries and religious groups. We exclude potential couples in which either partner is below the age of 10 from the analysis, considering these cases to be errors.

In the case of polygamous unions, the age of each female partner is subtracted from the age of the shared male partner (we did not encounter any cases of polyandry). Therefore, more than one age gap can be calculated for one man. Polygamy is rare in the vast majority of the 130 countries studied. However, according to our analysis, this arrangement is quite common throughout many countries in western Africa, and 11 per cent of people in sub-Saharan Africa live in polygamous households (in which multiple members are identified as the spouse or cohabiting partner of the same person). Countries with large shares of people living in polygamous households include Burkina Faso, where more than one-third of residents (36 per cent) live in a polygamous household, Mali (34 per cent), Nigeria (28 per cent), Senegal (23 per cent), Chad (15 per cent), and Ivory Coast (12 per cent). Polygamous partners are identified only when they share the same household.

At least 75 couples (150 people) are analysed for each country in our data set, and each religious group needs to be represented by the same sample size (at least 75 cases) for consideration at the country level. At the regional level, we set the sample size for a religious group at 200 cases, hence excluding Jews in Europe (80 cases) and Buddhists in sub-Saharan Africa (140 cases).

Within countries, sample weights are used. Partner age gap is an individual-level analysis. For social surveys, household surveys, and census data, we use the sample weights provided by the relevant survey organizations. The social survey sample weights we use are shown in Table 2. Household and IPUMS samples provide both person (i.e. individual) and household weights. We use person weights for analysing partner age gaps.

At the regional and global levels, we weight data using projected 2015 population sizes by country and religion (Hackett et al. 2015). For example, we weight each country's partner age gap by the number of people aged 15 and older and add the products of the weight and gap for each country together. We then divide the sum by the number of people aged 15 and older in the whole region.

### *Gender Inequality Index and per-capita GDP*

Two other measures we use are the Gender Inequality Index and per-capita gross domestic product (GDP). The Gender Inequality Index combines three different aspects of human development: reproductive health, women's empowerment, and

**Table 2** Weights used for each social survey

| Sample | Weight variable |
|---|---|
| European Social Survey | Pspweight |
| General Social Survey (US) | Wtss |
| Chinese General Social Survey | Weight |
| Korean General Social Survey | Weight |
| Taiwan Social Change Survey | wr_19_5 |

*Source:* 2010–18 survey data.

economic status; this measure was created by the United Nations Development Programme (UNDP 2019). The rationale for including gender inequality is that highly patriarchal countries tend to report larger spousal age gaps (Casterline et al. 1986).

Per-capita GDP values come from the *World Economic Outlook Database* (International Monetary Fund 2015) and are adjusted for purchasing power parity. The reason for including per-capita GDP is that higher levels of income are associated with smaller spousal age gaps (Atkinson and Glass 1985; Vera et al. 1985; Gustafson and Fransson 2015).

### *Singulate mean age at marriage*

We also consider differences in the SMAM by sex, using country-level estimates prepared by the United Nations (UN). SMAM values come from the UN's *World Marriage Data 2019* (United Nations 2019). For each country, we include the most recent data available between 2010 and 2018 in which the SMAM is disaggregated by sex. When given a choice between two sets of SMAM values for the same time period in a country, we use official national statistics rather than DHS data. We do not include countries whose most recent SMAM values for men and/or women are from before 2010.

There are 118 countries and territories with data available for both partner age gaps and SMAM differences. Twenty-four are in the Asia-Pacific region, 33 in Europe, 15 in Latin America and the Caribbean, eight in the MENA region, two in North America, and 36 in sub-Saharan Africa.

To calculate regional averages, we weight each country's SMAM difference by the number of people aged 15 and older and add the products of the weight and difference for each country together. We then divide the sum by the number of people aged 15 and older in the whole region.

## Results

### *Descriptive overview of partner age gaps*

Table 3 presents partner age gaps by religious group and region. Worldwide, men are an average of 4.2 years older than their wives or cohabiting female partners. However, there are large differences between regions. The largest partner age gap is seen among couples in sub-Saharan Africa (8.6 years), followed by those in the MENA region (6.1), the Asia-Pacific region (4.0), Latin America and the Caribbean (3.6), Europe (2.7), and North America (2.2). The country with the largest partner age gap is Gambia (14.5), while the country with the smallest gap is the Czech Republic (2.0): see Figure 1 for a map of partner age gaps by country and Table D1, supplementary material, for data on all 130 countries.

A few countries in Eastern Europe, such as Bulgaria (3.1 years), Hungary (3.0), and Romania (3.5), report larger partner age gaps compared with Europe as a whole (2.7). The same is true of countries in Southern Europe such as Albania (5.1), Croatia (3.6), Greece (4.3), Italy (3.4), Kosovo (4.1), Montenegro (4.9), North Macedonia (3.1), and Serbia (3.5). These countries tend to have lower per-capita GDPs and higher levels of gender inequality than nations in Western Europe.

Partner age gaps are larger in some religious groups than others. At a global level, the largest age gap is among Muslims (6.6 years), followed by Hindus (5.6), Christians (3.8), Buddhists (2.9), the religiously unaffiliated (2.3), and Jews (2.1). But the age gap of each religious group shows wide variability across regions. The partner age gap of Christians in sub-Saharan Africa is over three times as large as that of Christians in North America (7.1 vs 2.2 years), while the age gap for Muslims in sub-Saharan Africa is over twice as large as that of Muslims in Latin America and the Caribbean (11.6 vs 4.5).

**Table 3** Mean partner age gaps by world region and religious group, 2010–18

| | All | Christians | Muslims | Unaffiliated |
|---|---|---|---|---|
| Sub-Saharan Africa | 8.6 | 7.1 | 11.6 | 6.4 |
| Middle East and North Africa | 6.1 | 6.6 | 6.2 | – |
| Asia-Pacific | 4.0 | 3.0 | 5.7 | 2.2 |
| Latin America and the Caribbean | 3.6 | 3.6 | 4.5 | 3.8 |
| Europe | 2.7 | 2.6 | 5.0 | 2.4 |
| North America | 2.2 | 2.2 | – | 2.2 |
| World | 4.2 | 3.8 | 6.6 | 2.3 |

*Source:* As for Table 1.

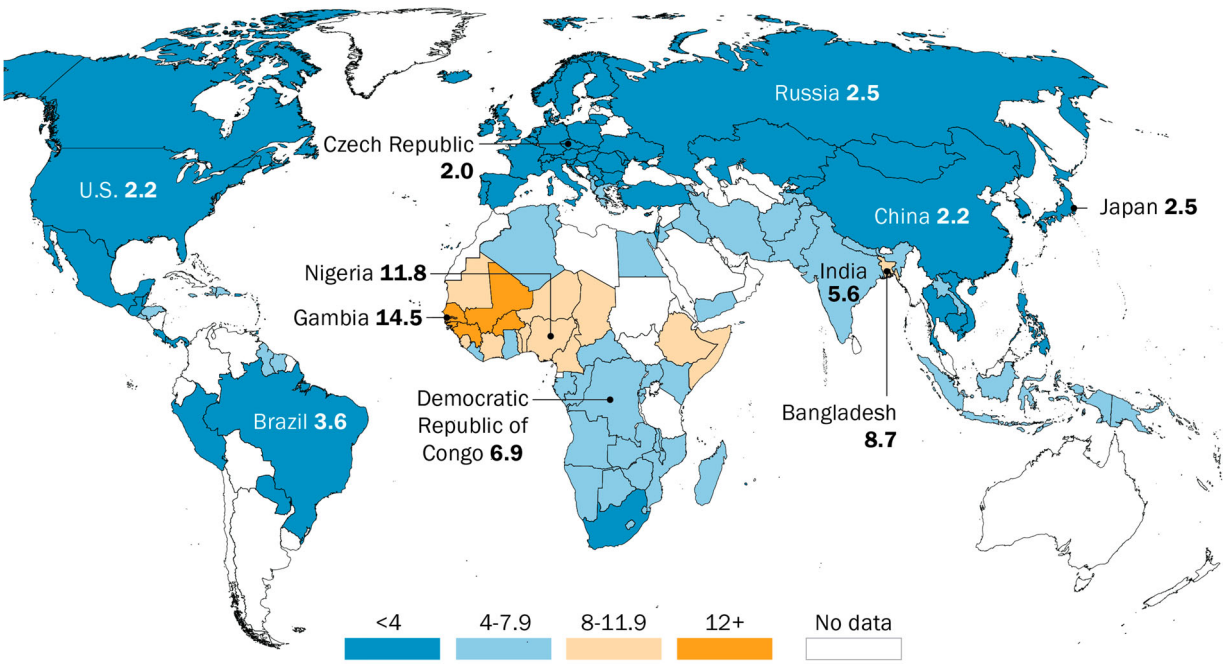


**Figure 1** Mean partner age gaps (in years) by country
*Source*: 2010–18 census and survey data; see Table A1, supplementary material for details.

Israeli Jews tend to show larger partner age gaps than American Jews (3.1 vs 1.0 years). The reason for this pattern is unclear, although it could be related to how Israeli Jews tend to be more religiously observant and more socially conservative on average compared with American Jews (Sahgal et al. 2016). This topic merits further research.

Even within individual countries, the size of the partner age gap often varies by religious group. For example, in Nigeria, Muslim men are 13.0 years older than their female partners on average, whereas Christian men are 9.2 years older than their partners.

### *Regression analysis*

Do religious differences in the partner age gap persist after accounting for country characteristics? To answer this question, we conduct an ordinary least squares (OLS) regression, with each country-level partner age gap by religion as the dependent variable and with independent variables for country-level per-capita GDP and gender inequality. The model also includes dummy variables classifying age gaps by couples' religion. An interaction term between gender inequality and per-capita GDP is included because these variables are highly correlated (r = −0.84). Results elsewhere in this paper are weighted by population, but for the purposes of this regression analysis only, country-level partner age gaps for each religious group with sufficient data are treated as equivalent units.

Table 4 presents *t*-statistics and *p*-values from the regression model. After controlling for the significant effects of gender inequality and per-capita GDP, differences between the partner age gaps of most religions are not statistically different from the religiously unaffiliated. However, age gaps are larger for Muslims than the unaffiliated ($t = 6.26$, $p < 0.001$). The model's adjusted R-squared value is 0.68.

### *Comparison with singulate mean age at marriage*

Table 5 compares partner age gaps with SMAM differences at regional and global levels. Globally, men are 4.2 years older on average than their female spouses or cohabiting partners (the partner age gap). However, the age difference between when men and women first marry is only 3.3 years (the SMAM difference).

The discrepancy between partner age gaps and SMAM differences is especially large in sub-Saharan Africa. In that region, men are 8.7 years older than their current female partners, on average, though they marry for the first time only 5.3 years later than women. Results are also striking

**Table 4** Linear regression model with country-level partner age gap by religion as the dependent variable and country-level gender inequality and per-capita GDP as independent variables

| | *t*-statistic | *p*-value |
|---|---|---|
| Gender Inequality Index (GII) | 14.53 | <0.001 |
| Per-capita GDP | 5.53 | <0.001 |
| Interaction between GII and per-capita GDP | −6.75 | <0.001 |
| Buddhist dummy | −0.66 | 0.511 |
| Christian dummy | 0.09 | 0.925 |
| Hindu dummy | −0.70 | 0.485 |
| Jewish dummy | 0.82 | 0.412 |
| Muslim dummy | 6.26 | <0.001 |
| Unaffiliated dummy | Omitted | |

*Source*: 2010–18 census and survey data; Gender Inequality Index values from the United Nations Development Programme (2019); per-capita GDP values from the International Monetary Fund (2015).

**Table 5** Comparison of mean partner age gap and SMAM difference, by world region, 2010–18

| | Partner age gap (years) | SMAM difference (men – women, years) |
|---|---|---|
| Sub-Saharan Africa | 8.7 | 5.3 |
| Middle East and North Africa | 6.1 | 4.7 |
| Asia-Pacific | 4.0 | 3.3 |
| Latin America and the Caribbean | 3.6 | 3.0 |
| Europe | 2.7 | 2.4 |
| North America | 2.2 | 1.7 |
| World | 4.2 | 3.3 |

*Notes*: In the analysis of 130 countries, men in sub-Saharan Africa are 8.6 years older than their female spouses or cohabiting partners on average. However, in this analysis of 118 countries, men in sub-Saharan Africa are 8.7 years older on average. All other partner age gaps by region are the same in both sets of analyses.
*Source*: 2010–18 census and survey data; *World Marriage Data* (United Nations 2019).

for the MENA region, where men are 6.1 years older than their female spouses or cohabiting partners, on average, but marry for the first time only 4.7 years later than women. By comparison, partner age gaps and SMAM differences are much closer in size in the Asia-Pacific region (4.0 vs 3.3 years), Latin America and the Caribbean (3.6 vs 3.0 years), Europe (2.7 vs 2.4 years), and North America (2.2 vs 1.7 years).

For country-level comparisons between the partner age gap and the SMAM difference, see Table D1 (supplementary material).

### *Comparison between government sources and social surveys*

We use social surveys to compute average partner age gaps for a few countries in which government censuses and demographic surveys lack data on religion. We would expect our country-level results to be similar if we had used censuses for these countries instead. Tables 6 and 7 compare average partner age gaps from government sources (American Community Survey for the US; census data extracted from IPUMS for European countries) with our results from selected social surveys. The differences in averages between government sources and social surveys range between zero and four months.

## Discussion

This study is noteworthy for several reasons. First, we focus primarily on age differences between current partners, as opposed to age differences when men and women first marry. This approach has not been used in many cross-national studies, with the notable exceptions of Casterline et al. (1986) and Barbieri et al. (2005). Our study updates and expands on these earlier works, which covered less than one-quarter as many countries as

**Table 6** Mean partner age gap in the US, based on General Social Survey (GSS) and American Community Survey (ACS), 2010–16

| | Partner age gap (years) | Difference from GSS (years) |
|---|---|---|
| GSS (pooled 2010, 2012, 2014, 2016) | 2.19 | N/A |
| ACS pooled five-year 2011–15 including cohabitators | 2.34 | 0.15 |
| ACS 2011 | 2.37 | 0.19 |
| ACS 2012 | 2.35 | 0.16 |
| ACS 2013 | 2.33 | 0.14 |
| ACS 2014 | 2.34 | 0.15 |
| ACS 2015 | 2.31 | 0.12 |
| ACS pooled five-year 2011–15 excluding cohabitators | 2.36 | 0.17 |

*Source*: 2010, 2012, 2014, 2016 General Social Survey; 2011–15 American Community Survey.

the present analysis. Other studies of married couples' age gaps have been limited to consideration of the SMAM (Saardchom and Lemaire 2005; Carmichael 2011; Abeynayake et al. 2012; Ortega 2014), mean age at first marriage (MacQuarrie 2016; OECD 2019), or median age at first marriage (Singh and Samara 1996).

An advantage of analysing age differences between all current partners is that it encompasses the realities of divorce, remarriage, and cohabitation. Spousal age gaps tend to widen when men and women enter their second (or later) marriages (Berardo et al. 1993; Wu et al. 2000). Another consideration is that polygamy is common in some countries in sub-Saharan Africa (Kramer et al. 2019). In polygamous households, age gaps are typically larger between men and their second wives than between men and their first wives (Tertilt 2003).

Our study also disaggregates results by religious group. Muslims stand out for their larger partner age gaps compared with non-Muslims, even after controlling for gender inequality and per-capita GDP. The reasons for this pattern are unclear, although it is possible that religious doctrines or customs could influence when Muslims marry. Most countries where polygamy is common have Muslim-majority populations and, within those countries, Muslims are usually more likely to practise polygamy than people from other religious groups (Kramer et al. 2019). Some Muslims who follow a conservative interpretation of the Quran argue that girls can get married on reaching puberty (Kohno et al. 2020). In addition, premarital sex is taboo in many Muslim communities, and this may encourage parents to seek marriages for their daughters at an early age (Adamczyk and Hayes 2012; Kohno et al. 2020).

Some previous cross-national age gap studies have included country-level religion variables (e.g. the share of adults in a country who are Muslim) in regression models (Garenne 2004; Saardchom and Lemaire 2005; Carmichael 2011; Abeynayake et al. 2012). However, our study is one of the first, along with MacQuarrie (2016), to report age gap statistics separately by religious group.

Our analysis is the most comprehensive and up-to-date cross-national study to look at age differences between current spouses or cohabiting partners. Our partner age gap data come from 130 countries that are home to 91 per cent of the global population. By comparison, Casterline et al. (1986) looked at age differences between spouses in 28 less developed countries, and Barbieri et al. (2005) focused on 18 countries in sub-Saharan Africa.

Similarly, many of the studies using the SMAM, mean age at first marriage, and median age at first marriage have been less comprehensive than ours, covering for example seven countries (MacQuarrie 2016), 32 countries (Garenne 2004), 39 countries (OECD 2019), 40 countries (Singh and Samara 1996), 66 countries (Abeynayake et al. 2012), and 77 countries (Carmichael 2011). Only two studies covered more: 156 countries (Saardchom and Lemaire 2005) and 227 countries and territories (Ortega 2014). Furthermore, many of these studies used data that are now quite old. For example, Saardchom and Lemaire (2005) analysed SMAM values from a 2000 UN data set, and Ortega (2014) analysed SMAM values that came primarily from a 2008 UN data set.

The results of our study line up nicely with conventional demographic wisdom. Countries with above-average levels of education and income, including nations in Europe and North America, report

**Table 7** Mean partner age gaps in European countries, based on European Social Survey and country censuses (IPUMS), 2010–18

| Country | ESS (years) | IPUMS (years) | Difference (IPUMS – ESS) | Note on IPUMS data | ESS year | IPUMS year |
|---|---|---|---|---|---|---|
| Austria | 3.0 | – | – | No household relationship variable | 2016 | 2011 |
| Belgium | 2.4 | – | – | Unavailable | 2016 | – |
| Bulgaria | 3.1 | – | – | Unavailable | 2012 | – |
| Croatia | 3.6 | – | – | Unavailable | 2010 | – |
| Cyprus | 3.1 | – | – | Unavailable | 2012 | – |
| Czech Republic | 2.0 | – | – | Unavailable | 2016 | – |
| Denmark | 2.8 | – | – | Persons not organized into households | 2014 | – |
| Estonia | 2.2 | – | – | Unavailable | 2016 | – |
| Finland | 2.2 | – | – | Persons not organized into households | 2016 | – |
| France | 2.3 | 2.535 | 0.274 | Rolling census, 31 per cent sample | 2016 | 2011 |
| Germany | 2.6 | – | – | Dated | 2016 | 1987 |
| Greece | 4.3 | 4.614 | 0.287 | Census, 10 per cent sample | 2010 | 2011 |
| Hungary | 3.0 | 2.986 | 0.010 | Census, 5 per cent sample | 2016 | 2011 |
| Iceland | 2.4 | – | – | Unavailable | 2016 | – |
| Israel | 3.4 | – | – | Dated | 2016 | 2008 |
| Italy | 3.4 | 3.209 | 0.142 | Census, 5 per cent sample | 2016 | 2011 |
| Lithuania | 2.3 | – | – | Unavailable | 2016 | – |
| Netherlands | 2.3 | – | – | Persons not organized into households | 2016 | – |
| Norway | 2.8 | – | – | Unavailable | 2016 | – |
| Poland | 2.6 | – | – | No household relationship variable | 2016 | 2011 |
| Russia | 2.5 | 2.186 | 0.335 | Census, 5 per cent sample | 2016 | 2010 |
| Slovakia | 2.1 | – | – | Unavailable | 2012 | – |
| Slovenia | 2.8 | – | – | Dated | 2016 | 2002 |
| Spain | 2.5 | – | – | No household relationship variable | 2016 | 2011 |
| Sweden | 2.5 | – | – | Unavailable | 2016 | – |
| Switzerland | 2.8 | – | – | Dated | 2016 | 2000 |
| Ukraine | 2.4 | – | – | Dated | 2012 | 2001 |
| United Kingdom | 2.4 | – | – | Dated | 2016 | 2001 |

*Source:* As for Table 1.

relatively small partner age gaps. Conversely, partner age gaps are larger in countries with high levels of gender inequality, including nations in the MENA region and in sub-Saharan Africa.

## Limitations

Like all studies, this one comes with some caveats. One limitation is that data quality may be negatively affected by methodological issues, such as age misreporting. Women responding to a survey may not know their own age or the age of their husband. As a result, they may sometimes give inaccurate information to an interviewer (Casterline et al. 1986; Van de Walle 1993). Another issue is that marital status may be misreported, particularly in countries where divorce and/or cohabitation are not socially acceptable (United Nations 2008).

Because religion data are not available for every household member, the household head or respondent's religion is applied to both members of each couple. While our analysis suggests that interfaith unions are relatively uncommon around the world, data limitations prevent us from measuring age gaps among interfaith couples across the countries in our study.

Couples in which neither member is the household head or respondent cannot be identified. For example, some couples who live within extended

families (which are common in some countries) cannot be included. Relationships between all household members are not recorded in rosters (which record only how members relate to the household head or respondent), so it is not possible to identify other couples. For example, if a survey respondent lives with her husband, son-in-law, and two daughters, only the age gap between the respondent and her husband can be extracted because it is unclear which daughter, if either, is the partner of the son-in-law.

We are missing partner age gap data from Gulf Cooperation Council (GCC) countries. However, in Saudi Arabia, for example, recent cohorts of Muslim women receive more education than co-religionists in less wealthy countries (McClendon et al. 2018). This may help to explain why the SMAM difference in Saudi Arabia (3.4 years) is not much different from the global average. Finally, this paper focuses on the male–female age gap between current partners and does not include information on people in same-sex partnerships or who do not identify as male or female.

## Future directions

Looking ahead, researchers could analyse differences between the partner age gaps of religious groups in more depth. Another potential future topic would be to compare the ages of married or cohabiting same-sex partners.

## Notes and acknowledgements

1 All four authors are based at the Pew Research Centre, Washington, DC, USA; Conrad Hackett is also at the University of Maryland, College Park, MD, USA.

2 Please direct all correspondence to Stephanie Kramer, 1615 L Street NW, Suite 800, Washington, D.C., 20036, USA; or by E-mail: skramer@pewresearch.org.

3 This paper relies on a global data set created for the Pew Research Center report, *Religion and Living Arrangements Around the World* (https://www.pewforum.org/2019/12/12/religion-and-living-arrangements-around-the-world/). Some of the results we report here are mentioned in that report and a related blog post. This work is part of the Pew-Templeton Global Religious Futures project, which is funded by The Pew Charitable Trusts and the John Templeton Foundation (Grant number 61059). The paper has benefited from the contributions of other Pew Research Center colleagues, including Alan Cooperman, Dalia Fahmy, and Yunping Tong.

## Disclosure statement

No potential conflict of interest was reported by the authors.

## ORCID

*Jacob Ausubel* http://orcid.org/0000-0001-6681-4961
*Stephanie Kramer* http://orcid.org/0000-0001-5263-7730
*Anne Fengyan Shi* http://orcid.org/0000-0003-1423-3924
*Conrad Hackett* http://orcid.org/0000-0002-1310-965X

## References

Abeynayake, Pamudi, Eduard J. Bomhoff, and Grace Lee. 2012. *Female Age at First Marriage and Fertility Levels: A Comparison of Developed and Developing Countries.* Paper presented at the 19th International Business Research Conference. https://doi.org/10.2139/ssrn.2174737

Adamczyk, Amy, and Brittany E. Hayes. 2012. Religion and sexual behaviors: Understanding the influence of Islamic cultures and religious affiliation for explaining sex outside of marriage, *American Sociological Review* 77(5): 723–746. https://doi.org/10.1177/0003122412458672

Allendorf, Keera, Arland Thornton, Colter Mitchell, Linda Young-DeMarco, and Dirgha J. Ghimire. 2017. Early women, late men: Timing attitudes and gender differences in marriage, *Journal of Marriage and Family* 79(5): 1478–1496. https://doi.org/10.1111/jomf.12426

Atkinson, Maxine P., and Becky L. Glass. 1985. Marital age heterogamy and homogamy, 1900 to 1980, *Journal of Marriage and the Family* 47(3): 685–691. https://doi.org/10.2307/352269

Barbieri, Magali, Véronique Hertrich, and Madeleine Grieve. 2005. Age difference between spouses and contraceptive practice in sub-Saharan Africa, *Population* 60(5): 617–654. https://doi.org/10.2307/4148187

BBC News. 2021. *Homosexuality: The Countries Where It Is Illegal to Be Gay.* https://www.bbc.com/news/world-43822234

Becker, Gary S. 1974. A theory of marriage, in Theodore W. Schultz (eds), *Economics of the Family: Marriage,*

*Children, and Human Capital*. Chicago, IL: University of Chicago Press, pp. 299–351.

Becker, Gary S. 1981. *A Treatise on the Family*. Cambridge, MA: Harvard University Press.

Berardo, Felix M., Jeffrey Appel, and Donna H. Berardo. 1993. Age dissimilar marriages: Review and assessment, *Journal of Aging Studies* 7(1): 93–106. https://doi.org/10.1016/0890-4065(93)90026-G

Caplan, David. 2017. *Emmanuel Macron's Wife on 25-Year Age Gap: 'We Have Breakfast Together, Me and My Wrinkles'*. New York, NY: ABC News.

Carmichael, Sarah. 2011. Marriage and power: Age at first marriage and spousal age gap in lesser developed countries, *The History of the Family* 16(4): 416–436. https://doi.org/10.1016/j.hisfam.2011.08.002

Casterline, John B., Lindy Williams, and Peter McDonald. 1986. The age difference between spouses: Variations among developing countries, *Population Studies* 40(3): 353–374. https://doi.org/10.1080/0032472031000142296

Conaway, Carrie. 2002. Chances aren't, *Regional Review – Federal Reserve Bank of Boston* 12(3): 24–30. https://ideas.repec.org/a/fip/fedbrr/y2002iq3p24-30nv.12no.3.html

Garenne, Michel. 2004. Age at marriage and modernization in sub-Saharan Africa, *Southern Africa Journal of Demography* 9(2): 59–79. https://journals.co.za/doi/pdf/10.10520/AJA16824482_104

Gustafson, Per, and Urban Fransson. 2015. Age differences between spouses: Sociodemographic variation and selection, *Marriage & Family Review* 51(7): 610–632. https://doi.org/10.1080/01494929.2015.1060289

Hackett, Conrad, Brian J. Grim, Marcin Stonawski, Vegard Skirbekk, Michaela Potančoková, and Guy Abel. 2012. *The Global Religious Landscape: A Report on the Size and Distribution of the World's Major Religious Groups as of 2010*. Washington, DC: Pew Research Center. https://www.pewforum.org/2012/12/18/global-religious-landscape-exec/

Hackett, Conrad. 2014. Seven things to consider when measuring religious identity, *Religion* 44(3): 396–413. https://doi.org/10.1080/0048721X.2014.903647

Hackett, Conrad, Phillip Connor, Marcin Stonawski, Vegard Skirbekk, Michaela Potančoková, and Guy Abel. 2015. *The Future of World Religions: Population Growth Projections, 2010-2050*. Washington, DC: Pew Research Center. https://www.pewforum.org/2015/04/02/religious-projections-2010-2050/

Hackett, Conrad, David McClendon, Michaela Potančoková, Marcin Stonawski, Caryle Murphy, and Vegard Skirbekk. 2016. *Religion and Education Around the World: Large Gaps in Education Persist, but all Faiths Are Making Gains – Particularly Among Women*. Washington, DC: Pew Research Center. https://www.pewforum.org/2016/12/13/religion-and-education-around-the-world/

Hackett, Conrad, Marcin Stonawski, Michaela Potančoková, Vegard Skirbekk, Phillip Connor, David McClendon, and Stephanie Kramer. 2017. *The Changing Global Religious Landscape*. Washington, DC: Pew Research Center. https://www.pewforum.org/2017/04/05/the-changing-global-religious-landscape/

Hackett, Conrad, Stephanie Kramer, Joey Marshall, Anne Fengyan Shi, and Dalia Fahmy. 2018. *The Age Gap in Religion Around the World*. Washington, DC: Pew Research Center. https://www.pewforum.org/2018/06/13/the-age-gap-in-religion-around-the-world/

Hackett, Conrad, Marcin Stonawski, Michaela Potančoková, Phillip Connor, Anne Fengyan Shi, Stephanie Kramer, and Joey Marshall. 2019. Projections of Europe's growing Muslim population under three migration scenarios, *Journal of Religion and Demography* 6: 87–122. https://doi.org/10.1163/2589742x-00601002

Hajnal, John. 1953. Age at marriage and proportions marrying, *Population Studies* 7(2): 111–136. https://doi.org/10.2307/2172028

International Monetary Fund. 2015. *World Economic Outlook Database*. October 2015 Edition. https://www.imf.org/en/Publications/WEO/weo-database/2015/October

Kohno, Ayako, Teeranee Techasrivichien, S. Pilar Suguimoto, Maznah Dahlui, Nik Daliana, Nik Farid, and Takeo Nakayama. 2020. Investigation of the key factors that influence the girls to enter into child marriage: A meta-synthesis of qualitative evidence, *PLoS ONE* 15(7): 1–20. https://doi.org/10.1371/journal.pone.0235959

Kramer, Stephanie, Dalia Fahmy, Anne Fengyan Shi, and Conrad Hackett. 2019. *Religion and Living Arrangements Around the World*. Washington, DC: Pew Research Center. https://www.pewforum.org/2019/12/12/religion-and-living-arrangements-around-the-world/

Livingston, Gretchen. 2014. *Typing the Knot Again? Chances Are, There's a Bigger Age Gap Than the First Time Around*. Washington, DC: Pew Research Center. https://www.pewresearch.org/fact-tank/2014/12/04/tying-the-knot-again-chances-are-theres-a-bigger-age-gap-than-the-first-time-around/

MacQuarrie, Kerry L. D. 2016. *Marriage and Fertility Dynamics: The Influence of Marriage Age on the Timing of First Birth and Birth Spacing*. Rockville, MD: Demographic and Health Surveys (DHS) Program. https://dhsprogram.com/publications/publication-as56-analytical-studies.cfm

McClendon, David, Conrad Hackett, Michaela Potančoková, Marcin Stonawski, and Vegard Skirbekk. 2018. Women's education in the Muslim world, *Population and Development Review* 44(2): 311–342. https://doi.org/10.1111/padr.12142

Organisation for Economic Co-operation and Development. 2011. Families are changing, in Organisation for Economic Co-operation and Development, *Doing Better for Families*. pp. 17–53. https://www.oecd.org/els/soc/47701118.pdf

Organisation for Economic Co-operation and Development. 2019. *SF3.1: Marriage and Divorce Rates*. Paris, France: Social Policy Division. https://www.oecd.org/els/family/SF_3_1_Marriage_and_divorce_rates.pdf

Ortega, José A. 2014. A characterization of world union patterns at the national and regional level, *Population Research and Policy Review* 33(2): 161–188. https://doi.org/10.1007/s11113-013-9301-x

Saardchom, Narumon, and Jean Lemaire. 2005. Causes of increasing ages at marriage: An international regression study, *Marriage & Family Review* 37(3): 73–97. https://doi.org/10.1300/J002v37n03_05

Sahgal, Neha, Alan Cooperman, Angelina E. Theodorou, Katie Simmons, Kelsey Jo Starr, Steve Schwarzer, Besheer Mohamed, et al. 2016. *Israel's Religiously Divided Society*. Washington, DC: Pew Research Center. https://www.pewresearch.org/religion/2016/03/08/israels-religiously-divided-society/

Shehan, Constance L., Felix M. Berardo, Hernan Vera, and Sylvia M. Carley. 1991. Women in age-discrepant marriages, *Journal of Family Issues* 12(3): 291–305. https://doi.org/10.1177/019251391012003003

Singh, Susheela, and Renee Samara. 1996. Early marriage among women in developing countries, *International Family Planning Perspectives* 22(4): 148–157. https://doi.org/10.2307/2950812

Tertilt, Michèle. 2003. Polygyny and poverty. Research Paper, Stanford University Department of Economics. https://economics.sas.upenn.edu/sites/default/files/filevault/event_papers/Empirical040820040.pdf

United Nations Development Programme. 2019. *Human Development Data (1990–2018)*. http://hdr.undp.org/en/data

United Nations Population Fund. 2020. *Child Marriage – Frequently Asked Questions*. https://www.unfpa.org/child-marriage-frequently-asked-questions

United Nations. 1983. *Manual 10: Indirect Techniques for Demographic Estimation*. New York, NY: Department of Economic and Social Affairs, Population Division. https://www.un.org/en/development/desa/population/publications/pdf/mortality/Manual_X.pdf

United Nations. 2008. *Singulate Mean Age at Marriage*. New York, NY: Department of Economic and Social Affairs, Population Division. https://www.un.org/en/development/desa/population/publications/dataset/marriage/age-marriage.asp

United Nations. 2013. *World Fertility Report 2012*. New York, NY: Department of Economic and Social Affairs, Department of Economic and Social Affairs, Population Division. https://www.un.org/en/development/desa/population/publications/dataset/fertility/wfr2012/Metadata/Metadata_MAFM-SMAM-EVER-MARRIED.pdf

United Nations. 2016. *Changing Patterns of Marriage and Unions Across the World*. New York, NY: Department of Economic and Social Affairs, Population Division. https://www.un.org/en/development/desa/population/publications/pdf/popfacts/PopFacts_2016-2.pdf

United Nations. 2019. *Progress of the World's Women 2019-2020: Families in a Changing World*. United Nations Entity for Gender Equality and the Empowerment of Women. https://www.unwomen.org/sites/default/files/Headquarters/Attachments/Sections/Library/Publications/2019/Progress-of-the-worlds-women-2019-2020-en.pdf

United Nations Population Division. 2019. *World Marriage Data 2019*. https://www.un.org/development/desa/pd/data/world-marriage-data

Van de Walle, Etienne. 1993. Recent trends in marriage ages, in Karen A. Foote, Kenneth H. Hill and Linda G. Martin (eds), *Demographic Change in Sub-Saharan Africa*. Washington, DC: National Academy Press, pp. 117–152.

Vera, Hernan, Donna H. Berardo, and Felix M. Berardo. 1985. Age heterogamy in marriage, *Journal of Marriage and the Family* 47(3): 553–566. https://doi.org/10.2307/352258

Wodon, Quentin. 2015. Child marriage, family law, and religion: An introduction to the fall 2015 issue, *The Review of Faith & International Affairs* 13(3): 1–5. https://doi.org/10.1080/15570274.2015.1075761

Wu, Zheng, Karli H. Burch, Randy Hart, and Jean E. Veeevers. 2000. Age-Heterogamy and Canadian unions, *Social Biology* 47(3): 277–293. doi:10.1080/19485565.2000.9989023